Kuijun Liu

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Education

2022–present	**Rutgers University** Ph.D. in Mathematics, expected May 2027 Advisor: Feng Luo
2022	**Université Paris-Saclay** Master (M2) in Statistics and Machine Learning, mention Bien
2021	**Imperial College London** M.Sc. in Applied Mathematics, with Distinction
2020	**University of Liverpool / Xi'an Jiaotong-Liverpool University** B.Sc. in Mathematics (2+2 program), First Class Honours

Research Interests

Discrete differential geometry (discrete conformal geometry, rigidity of polyhedral surfaces); hyperbolic geometry; computational geometry (nearest-neighbor search).

Papers and Manuscripts

1. **The Liouville property for geometric graphs in the plane** arXiv:2610.39242
   Joint with Feng Luo, Yanwen Luo, and Zhenghao Rao.

   A Liouville theorem for the cotangent Laplacian: every bounded discrete harmonic function on a Delaunay triangulation of the Euclidean plane is constant; the main theorem establishes this for a more general class of connected weighted geometric graphs in the plane. Cotangent-weighted Delaunay triangulations of the plane can be transient, but are recurrent when edge lengths are uniformly bounded.

2. **Discrete Schwarz lemmas for convex caps** *in preparation*
   Joint with Feng Luo, Yanwen Luo, and Zhenghao Rao.

   A Euclidean discrete Schwarz lemma: for admissible convex caps with a fixed intrinsic roof metric, ordered boundary data and interior curvature give a comparison of vertex heights. Extensions to the hyperideal setting via admissible heights, curvature rigidity, and comparison principles; regularity of the Hilbert–Einstein functional across triangulation changes, via edge flips and geometric height estimates.

3. **Navigable Graphs for Low-Dimensional Hyperbolic Spaces** arXiv:2610.24874
   Joint with Chengyuan Deng, Jie Gao, Kevin Lu, Feng Luo, Puheng Weng, and Cheng Xin.
   Submitted to ITCS 2027.

   Sparse graphs for greedy routing in fixed-dimensional hyperbolic spaces, combining local navigation with a global geometric hierarchy. Angle and distance estimates in the Poincaré model for cone-based neighbor selection give the local routing guarantees and an approximate nearest-neighbor analysis.

Conferences and Summer Schools

Jul.–Aug. 2025	*Topological and Geometric Structures in Low Dimensions*, SLMath summer school
Feb. 2025	*Circle Packings, Minimal Surfaces, and Discrete Differential Geometry*, ICERM workshop
Jun. 2024	*Special Geometric Structures and Analysis*, SLMath summer school
2024, 2025	Graduate Student Conference in Algebra, Geometry, and Topology, Temple University

Teaching and Mentoring

Instructor of record, Rutgers University

Spring 2026	Advanced Calculus for Engineering (Math 421)
Summer 2024	Calculus I (Math 151)

Teaching assistant and grader, Rutgers University

Fall 2026	Linear Algebra (Math 250), TA at large
Fall 2025	Calculus II (Math 152)
Spring 2025	Linear Algebra (Math 250), TA at large
Fall 2024	Introduction to Real Analysis I (Math 311)
Spring 2024	Multivariable Calculus (Math 251)
Fall 2023	Calculus II (Math 152)
2022–2023	Grader for Introduction to Real Analysis (Math 311, two semesters), Linear Algebra (Math 350), and Multivariable Calculus (Math 251)

Mentoring

Summer 2025	REU co-supervisor, project on inversive circle packing

Other Research Experience

Apr.–Sep. 2022	**CNRS–IBENS, Paris**, M2 research internship (supervisor: David Holcman) Real-time detection of spike-and-wave discharges in EEG, using wavelet denoising, signal decomposition, and SVM classifiers.